

Estimating spatial firm productivity spillovers using UK microdata

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1. LITERATURE

2. DATA

2.1. FAME

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular, number of employees and balance sheet data from which analysis-relevant aggregates can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey⁴.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim for (Irastorza-Fadrique et al., 2026). and reused for this project. Raw data is provided on a series of a financial balance sheet indicators, including characteristics that are fixed for each firm, bucketed into *a1_ID* and *a5_misc* indicator tables, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

2.2. INGESTION AND CLEANING

(...)

All data is deflated to real terms, using 2024 as the base year. I used the GDP deflator for all balance sheet variables, except for *renumeration_employees* and its composites for which I used CPIH, as these are income-related aggregates with real spending power best captured by household purchasing baskets⁵.

All balance sheet data are in thousands ('000s) of GBP (£). As we estimate log relationships exclusively, this has no effect on parameter interpretation.

2.3. LOCATION DATA

2.4. DERIVING PANEL AGGREGATES

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame's **fixed_total** variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for

⁴?

⁵for National Statistics (2026)

positive values Table (I), which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE I
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratios
tangibles		3.53
fixed_other	tangibles + intangibles + inv._other	6.00
current_assets	fixed_total + current_assets	11.56

GVA is calculated as the sum of *ebitda* and *renumeration_employees*. *renumeration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via the sum of *profit_loss_pretax*, *interest_paid*, and *depreciation*, providing an alternative measure of GVA. We can expect the difference between these two to be accounting differences⁶ and additionally note that the alternate GVA measure also reduces panel observations from 1,085,520 to 618,588. For these reasons, I do not report TFP calculations with this second GVA measure which are regardless near-identical, though I resultingly refer to GVA throughout this paper as *gva_1*.

2.5. FIRM DISTANCE MEASURES

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. This relationship is irreflexive. For groups it is transitive: if firms A and B share a spatial group, such as a travel-to-work area (*ttwa*), first-half of a UK postcode (*pc4*), or full postcode (*pc8*), then a firm C that is a peer of firm B must also be of firm A. For continuous distance measures, this is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are not shared with A. Every firm is then its own overlapping group, or equivalently, a sparse, rather than block diagonal, firm-to-firm matrix⁷.

(Cleaning address data).

Fame only reports each company’s last-known registered and trading address, at the time of compilation or publication of its annual vintage(cite). As a result, final location data is only available at the firm level. We can interpret each firm as being static over the entire 20-year sample, with no movers. This seems very unlikely (expand). Instead, we must interpret the constructing address, and therefore firm-to-firm distance measures as a signal of firm pairs’ true distance polluted by movers. This has 2 econometric consequences.

Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, their firm location decision is endogenous to the subsequent productivity spillovers they experience and exert, for reasons discussed in Section 1 on New Economic Geography. (More on firm FEs) There is no solution to this for moves occurring out-of-sample, for instance, just before 2009.

Secondly, I acknowledge some attenuation to the productivity spillover estimate Peer firms

⁶An OLS regression between the two, discarding 5% outliers, has a coefficient of 0.91 with a 0.000 p-value. This means *gva_2* is slightly lower than *gva_1*, with that impact mostly driven by top-end outliers.

⁷I do actually impose a block diagonal structure for reasons of computation as explained subsequently

who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location (More on direction of bias/positive selection?)

I first consider a model where each target has peers which share a spatial identifier, meaning they belong to the same travel-to-work-area (*ttwa*) using the 2011 definitions ⁸, the first half of a UK postcode *pc4*, and the full UK postcode *pc8*.

(Discuss and illustrate different forms of ω , with illustrative block diagonal matrices)

⁸(Review ONS bulletin on TTWAs data source and describe clearly)

3. MODEL

I use a stylised, basic 3-factor production model throughout this analysis. Y , L , K are GVA, labour, and capital as usual. Z_{jt} is the 2-factor total-factor productivity (TFP) residual. I use lower-case variables to indicate logged terms and bolded variables to indicate stacked vectors.

$$Y_{it} = A \cdot L_{it}^\alpha K_{it}^\beta \left[\prod_{j \neq i} f(Z_{jt}) \right]^\gamma \quad (1)$$

$$y_{it} = c + \alpha l_{it} + \beta k_{it} + \gamma \sum_{j \neq i} \ln f(Z_{jt}) \quad (2)$$

$$z_{it} = y_{it} - c - \alpha l_{it} - \beta k_{it} \quad (3)$$

$$f(Z_{jt}) = Z_{jt}^{w_{ijt}} \quad (4)$$

$$z_{it} = c + \gamma \sum_{j \neq i} w_{ijt} z_{jt} \quad (5)$$

$$\mathbf{z}_t = c + \gamma \boldsymbol{\Omega}_t \mathbf{z}_t \quad (6)$$

Theory suggests no restrictions on α , β , and γ collectively should be imposed, as γ is a positive externality. (Lucas discusses this)

In equation 4 I assume log-linearity for the peer TFP effect on the target, namely the $f(\cdot)$ transformation function dictating effects for peers contemporaneously is a Cobb-Douglas weighted set of individuals, s.t. $\sum_{j \neq i} w_{ijt} = 1$. This permits me to estimate structural γ in two stages should I require, namely using y_{it} log GVA as the regressand is equivalent to first estimating z_{it} via a 2-factor TFP estimation and then secondly regressing z_{it} against its peers.

To estimate structural γ , I use 3 linear models, increasing in complexity, each resolving some of the endogeneity and identification issues of the last. These are summarised in Table II below. Given that only the last, Section 3.4 should be interpreted as causal, we could have skipped straight to analysis with this model and omitted the rest, namely Sections 3.2 and 3.4. However, these incremental steps are useful for iteratively understanding the data and making choices about parameter selection along the way and are computationally easier to calculate, allowing more parameter examinations to be undertaken, assuming that the relationships are similar between these parameter choices household in both the non-causal and causal specifications. Choosing all these parameters once, in a final model, would have required a large amount of computation via cross-validation, which this investigations limits on time and computation did not permit.

TABLE II
(Summary of econometric models)

Model	Stages	Identification	Endog.	Consistent?	Decisions
1. LMM grouped peers	2				
2. Static distance decay	2				
3. Dynamic panel	1				

'Stages' indicates whether the derived 2-factor TFP value z_{it} was used as the regressand, or the model used y_{it} from the *gval* variable directly.

3.1. 2-FACTOR TFP ESTIMATION

As established above, it is helpful to conduct a first stage derividing the Solow residual, z_{ijt} , from a 2-factor production model, which I will analyse and decompose further in a second stage. This measurement is only helpful for static models, so the spillover from peer firms via Z_{jt} only occurs in the same t -period. As I will show, there may be a dynamic lag from peer TFP z_{jt} to target TFP z_{it} , so for the later model, z_{it} needs to be estimated in full in a single-stage with dynamic variables.

$$\begin{aligned} y_{it} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 l_{it} + \beta_2 k_{it} + \varepsilon_{it} \\ z_{it} &= y_{it} - c - \alpha l_{it} - \beta k_{it} \end{aligned} \tag{7}$$

I conduct the reduced-form static panel regression in logs, with firm and time fixed effects. I assume exogeneity between the residual and factor input levels.

TABLE III
Baseline TFP calculation from 2-factor production

Parameter	gva1_ft	gva2_ft	k2_ft	gva1_f	gva1_t	gva1_
β_0	3.201	3.445	4.840	3.223	1.994	3.201
const	(0.032)***	(0.051)***	(0.019)***	(0.027)***	(0.010)***	(0.032)***
β_1	0.621	0.624	0.717	0.618	0.559	0.621
ln_employees	(0.005)***	(0.007)***	(0.005)***	(0.004)***	(0.002)***	(0.005)***
β_2			0.058			
ln_fixed_total			(0.001)***			
β_3	0.265	0.245		0.264	0.422	0.265
ln_total_assets	(0.004)***	(0.007)***		(0.004)***	(0.002)***	(0.004)***
Observations	1,081,520	618,588	1,042,854	1,081,520	1,081,520	1,081,520
i	140,458	93,041	135,428	140,458		140,458
t	19	19	19		19	19
$R^2(within)$	0.386	0.323	0.339	0.386	0.366	0.386

All parameters reported with standard errors underneath. Significant estimates against an $H_0 : \beta = 0$ are indicated with $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***).

The parameter estimates give a labour income share exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share* which is constructed from a similar sample to the coverage in Fame⁹. Specifications without fixed or time effects offer similar parameter estimates. The capital share of income jumps on a specification with only time fixed effects, which is to be expected (explain). Oddly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income from the regression. For the rest of the analysis, I use *total_assets* as the primary capital measure and leave it for future analysis why this dataset yields this phenomenon.

TFP is the residual on GVA after extracting the proportion that can be accounted by capital and labour inputs. Equivalently, it is the sum of fixed effects in this specification with the error term, which means we should expect it to be exogenous to capital and labour inputs,

⁹for National Statistics (2024)

but correlated to both firm effects and time trends. By extracting this regression residual, I have a TFP measure for all of my 1,081,520 panel observations.

3.2. LMM GROUPED PEERS

Section 2.5 discusses how firms have been labelled to symmetric, transitive, non-overlapping groups by *pc8* (smallest), *pc4*, and *ttwa* (largest). I conduct a linear regression using the averages of peers belonging to these groups, leaving out the target firm. This is known as the 'Linear-in-Means (LMM)' Model and is common to the peer effects literature.

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \sum_{g=1}^3 \beta_g \mathbb{E}[z_{jtg}|g] + \epsilon_{it} \quad (8)$$

$$\mathbf{z}_t = c + \gamma \mathbf{\Omega}_t \mathbf{z}_t$$

$$w_{ij} \in \mathbf{\Omega}_t = \begin{cases} 1, & \text{if } (i, j) \in g \\ 0, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

It is crucial that I have multiple, non-overlapping groups in a single model, where my target firm z_{it} is a member of all groups simultaneously, but peers in one group are excluded from all others.

The non-overlapping property prevents multicollinearity. If I were to use nested, overlapping geographies, namely keeping all firms within the *ttwa* area in that group average, including the ones I have just counted in the *pc4* and *itxpc8* groups, then my group averages mechanically align: $z_{jt,pc8}$ is embedded within $z_{jt,pc4}$, and both are within $z_{jt,ttwa}$, making β_g parameter estimates unstable.

The single model approach prevents omitted variable bias (OVB). $\mathbb{E}[z_{jt,ttwa}|g] \propto \mathbb{E}[z_{jt,pc8}|g]$, so running three models where the former is just my single regressor means the latent $\beta_{pc8} \in u_{it}$ is correlated with both z_{itg} and z_{jtg} . We could reasonably argue that the inner rings are causal, so this causal factor excluded from the regression influences both and creates heavily upward bias on my outer-ring estimations. Indeed, this critique extends within further inner rings. If I think the causal unit is not peer firms within the *pc8* group at all, but a further hyper-local effect - perhaps those on the same street or same building, then this latent factor will correlate with both my outcome and regressor and drive the results. However, given the small number of peers contained within an *pc8* unit and the limits of data availability to identify firms at more local granularities, I proceed assuming the effect from *pc8* peers is homogenous.

Finally, the approach using multiple groups as regressors at once gives identification by breaking transitivity. If I were to run the same regression with merely one group, say *pc8*, then my regression would suffer the reflection problem outlined in (Manski, 1993). As I am estimating the expected value of z_{it} and z_{jt} from the same sample, they cannot be uniquely identified in reduced form.

Reflection problem with a single regressor, ι denotes a column vector of 1s

(review: maths is a bit tricky here)

$$\begin{aligned}
\mathbb{E}[z_{itg}|g] &= \alpha_i + \gamma_t + \beta_0 + \beta_1 \mathbb{E}[z_{jtg}|g] \\
&= (\alpha_i + \gamma_t + \beta_0)\iota + \beta_1 \mathbf{\Omega}_t \mathbb{E}[\mathbf{z}_{tg}|g] \\
&= (\mathbb{I} - \beta_1 \mathbf{\Omega}_t)^{-1}(\alpha_i + \gamma_t + \beta_0)\iota
\end{aligned}$$

This problem is mitigated with having 3 distinct groups to which z_{it} belongs. The results of the LMM estimation implementing equation 8 are below:

TABLE IV
Static panel linear-in-means using spatial-group TFP averages

Parameter	lmm	lmm_no_firm_fe	lmm_employees
β_0	2.075	0.310	1.202
const	(0.044)***	(0.038)***	(0.215)***
β_1	0.033	0.204	0.306
peer_tfp_ttw_a_donut	(0.013)***	(0.013)***	(0.071)***
β_2	0.015	0.142	0.015
peer_tfp_pc4_donut	(0.005)***	(0.008)***	(0.005)***
β_3	0.280	0.559	0.280
peer_tfp_pc8	(0.005)***	(0.004)***	(0.005)***
β_4			-0.058
peer_tfp_pc8_x_employees			(0.015)***
β_5			0.184
ln_employees			(0.044)***
Observations	764,537	764,537	764,537
i	107,143		107,143
t	19	19	19
$R^2(within)$	0.040	-0.000	0.040

(Reformat this table: reorder variables, rename, remove no fe)

The parameter results suggest a strong correlation between the TFP of a target firm and its peers. Each additional outward group only adds to explain the causal variation

Crucially however, these parameter estimates should not be interpreted as causal. (thinking about non-overlapping groups theory) (... reflection problem)

3.3. STATIC DISTANCE DECAY

$$y_{it} = \alpha_i + \gamma_t + \beta_0 + \beta_1 l_{it} + \theta w_{it} + \epsilon_{it} \quad (10)$$

$$z_{it} = \alpha_i + \gamma_t + \theta w_{it} + \beta f(z_{jt}) + \epsilon_{it} \quad (11)$$

TABLE V
Static panel using continuous distance-decay weighted TFP average of peers

Parameter	dd_gravity1	dd_neg2	dd_expdd3	dd_struct1	dd_struct2	dd_struct3
β_0	2.479	2.586	2.417	2.698	2.799	2.652
const	(0.029)***	(0.029)***	(0.032)***	(0.033)***	(0.032)***	(0.035)***
β_1	0.168			0.168		
tfp_wav1	(0.003)***			(0.003)***		
β_2		0.135			0.135	
tfp_wav2		(0.003)***			(0.003)***	
β_3			0.184			0.183
tfp_wav3			(0.005)***			(0.005)***
β_4	-0.009	-0.009	-0.009	0.619	0.619	0.619
ln_employees	(0.004)**	(0.004)**	(0.004)**	(0.005)***	(0.005)***	(0.005)***
β_5	0.013	0.012	0.016			
ln_nb_peers	(0.004)***	(0.004)***	(0.004)***			
β_6				0.265	0.265	0.265
ln_total_assets				(0.004)***	(0.004)***	(0.004)***
Observations	1,067,013	1,067,013	1,067,013	1,067,013	1,067,013	1,067,013
i	139,162	139,162	139,162	139,162	139,162	139,162
t	19	19	19	19	19	19
$R^2(within)$	0.010	0.010	0.004	0.392	0.391	0.388

3.4. DYNAMIC PANEL

TABLE VI
Dynamic panel using instrumented continuous
distance-decay weighted TFP

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	1.371 (0.134)***	1.577 (0.144)***	1.064 (0.103)***
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
$\ln(employees)$	0.696 (0.038)***	0.689 (0.037)***	0.709 (0.040)***
$\ln(total_assets)$	0.259 (0.041)***	0.266 (0.041)***	0.247 (0.044)***
tfp_wav_1	0.503 (0.028)***		
tfp_wav_2		0.423 (0.023)***	
tfp_wav_3			0.618 (0.042)***
Hansen Test	X (0.417)	X (0.459)	X (0.246)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.746)	X (0.600)	X (0.945)
i	139,162	139,162	139,162
Observations	713,339	713,339	713,339

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APPENDIX 1

APPENDIX 2

APPENDIX 3